# ACR-VLA: Inference-Time Dimension-Selective Action Correction via Multivariate Conformal Prediction for Vision-Language-Action Models

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**Abstract**

Flow-matching Vision-Language-Action (VLA) models achieve near-saturated performance on standard manipulation benchmarks yet suffer catastrophic robustness failures under distribution shift. We present **ACR-VLA** (Adaptive Conformal Refinement for VLAs), an inference-time module that extracts per-degree-of-freedom (per-DOF) conformal statistics from multiple flow-matching samples and feeds them into a lightweight Conformal Refinement Module (CRM) to perform *dimension-selective* action correction. Three components constitute the pipeline: (i) **Conformal Geometry Statistics (CGS)**, which computes 45-dimensional structured uncertainty features encoding per-DOF standard deviations, conformal radii, and ellipsoid volumes from $K$=10 samples with Ledoit–Wolf shrinkage; (ii) **Adaptive Conformal Inference (ACI)** driven by a novel **Chunk Overlap Consistency (COC)** signal that maintains calibrated coverage without external supervision; and (iii) a **CRM** (131K parameters, $\sim$12 ms overhead) that outputs a learned residual correction conditioned on the CGS features. On LIBERO-Plus, ACR-VLA improves a base SmolVLA from 57.8% to 72.8% success rate (+15.0 pp) while preserving nominal LIBERO performance (96.5% vs. 97.3%). When stacked on SRPO—the current training-time state of the art at 82.1%—ACR-VLA achieves **88.7%**, establishing a new LIBERO-Plus SOTA and demonstrating that inference-time conformal correction provides orthogonal gains to training-time methods. Ablations confirm that per-DOF conformal features are the primary driver (12.0 pp gap over generic residual), and that the COC self-referential signal enables 2.5$\times$ faster ACI convergence than episode-level supervision.

## 1 Introduction

Flow-matching Vision-Language-Action (VLA) models have achieved remarkable success on standard robotic manipulation benchmarks, with top systems reaching 97–99% success rates on LIBERO [1]. However, LIBERO-Plus [2] reveals a sobering reality: these same models collapse to 50–60% success under realistic distribution shifts including camera viewpoint changes, sensor noise, and lighting variations.

Existing approaches to VLA robustness fall into two categories. *Training-time methods* such as SRPO [3], RobustVLA, and DiG-Flow improve the base policy through data augmentation or reinforcement learning, achieving up to 82.1% on LIBERO-Plus. *Inference-time methods* such as ReconVLA [4] and SAFE [5] use uncertainty estimates for sample selection or anomaly detection, but rely on scalar uncertainty scores that discard the rich per-DOF covariance structure available from flow-matching samples.

We identify a critical gap: no existing method leverages *structured, per-DOF conformal statistics* as input features for a learned correction module. ReconVLA uses scalar Euclidean distance for sample selection—it cannot synthesize an action better than any individual sample. A2C2-style generic residual policies lack calibrated uncertainty conditioning. CP-in-the-Loop employs conformal prediction for planning but not as features for learned correction.

**ACR-VLA** bridges this gap with three contributions:

1. **Per-DOF conformal features for dimension-selective correction.** We show that 45-dimensional CGS features (per-DOF std, conformal radius, and ellipsoid volume) enable a lightweight CRM to se-

lectively correct high-uncertainty DOFs while preserving confident predictions. Ablations demonstrate a 12.0 pp gap over CRM without conformal features and a 6.0 pp gap over scalar uncertainty conditioning.
2. **COC as a self-referential ACI signal.** Chunk Overlap Consistency—a measure of inter-chunk prediction agreement—correlates with task success ($\rho$=0.61, $p<10^{-4}$) and enables ACI convergence in 47 episodes vs. 120 for episode-level supervision, without requiring external reward signals.
3. **Orthogonal gains to training-time methods.** ACR-VLA stacked on SRPO yields 88.7% on LIBERO-Plus (+6.6 pp over SRPO alone), demonstrating that inference-time conformal correction and training-time RL optimization address complementary failure modes.

# 2 Related Work

**Robustness of VLA models.** LIBERO-Plus exposes the fragility of VLAs under distribution shift. Training-time approaches include domain randomization (RobustVLA), flow-matching augmentation (DiG-Flow), and RL fine-tuning (SRPO, 82.1% LIBERO-Plus SOTA). Inference-time methods include ReconVLA (scalar uncertainty-based sample selection) and SAFE (anomaly detection for safe fallback). ACR-VLA is the first to combine multivariate conformal prediction with learned residual correction at inference time.

**Conformal prediction in robotics.** Conformal prediction provides distribution-free coverage guarantees. Recent work applies CP to robot planning (CP-in-the-Loop), policy gating (UPS), and safe set construction. These methods treat CP outputs as binary decisions (inside/outside the conformal set) rather than as continuous features for downstream learning. ACR-VLA instead feeds per-DOF conformal statistics directly into a learned correction module, enabling graded, dimension-selective responses.

**Uncertainty quantification for action correction.** DiU estimates aleatoric uncertainty via a single forward pass; A2C2 learns generic residual corrections. Neither provides calibrated, per-DOF epistemic uncertainty features. ACR-VLA's CGS captures epistemic uncertainty (elevated under distribution shift, low in-distribution), which we show is fundamentally more informative for action correction than aleatoric estimates (Section 4.8).

# 3 Method

ACR-VLA operates as a post-hoc inference-time module on top of any flow-matching VLA. Given a base VLA $\pi_\theta$ that generates action chunks via flow matching, ACR-VLA draws $K$=10 samples, computes structured uncertainty features, and produces a corrected action. The pipeline consists of three components: CGS (§3.1), ACI+COC (§3.2), and CRM (§3.3).

## 3.1 Conformal Geometry Statistics (CGS)

Given $K$ action samples $\{a^{(k)}\}_{k=1}^{K}$ from the base VLA, where each $a^{(k)} \in \mathbb{R}^d$ ($d$=7 DOFs for a typical manipulator: 3 translation + 3 rotation + 1 gripper), CGS computes:

**Sample statistics.** The sample mean $\bar{a} = \frac{1}{K}\sum_k a^{(k)}$ and covariance $\hat{\Sigma} = \mathrm{LW}(\{a^{(k)}\})$, where LW denotes Ledoit–Wolf shrinkage. Shrinkage is essential: at $K$=10 and $d$=7, the ratio $p/n = 0.7$ makes the raw sample covariance ill-conditioned. Ledoit–Wolf reduces the condition number from $\sim 10^4$ to $\sim 10^2$.

**Per-DOF features.** For each DOF $j \in \{1, \ldots, d\}$, CGS extracts: (i) marginal standard deviation $\sigma_j = \sqrt{\hat{\Sigma}_{jj}}$; (ii) conformal radius $r_j$, defined as the $j$-th marginal projection of the Mahalanobis ellipsoid at the $(1-\alpha)$ quantile; and (iii) a shared ellipsoid volume $V = \det(\hat{\Sigma})^{1/2}$. This yields a 15-dimensional feature vector per DOF (for a 7-DOF action chunk spanning 2–3 timesteps) plus the scalar volume, totaling 45 dimensions.

**Nonconformity score.** The joint Mahalanobis distance $s(a) = (a-\bar{a})^\top \hat{\Sigma}^{-1}(a-\bar{a})$ serves as the nonconformity score for conformal calibration. This captures cross-dimensional correlations that per-DOF Bonferroni correction misses (Section 4.7).

### 3.2 Adaptive Conformal Inference with COC

**Chunk Overlap Consistency (COC).** Flow-matching VLAs predict action chunks spanning multiple future timesteps. Consecutive predictions overlap in their temporal coverage. COC measures the agreement between overlapping portions of adjacent chunks:

$$\mathrm{COC}_t = 1 - \frac{\|\hat{a}_t^{\text{overlap}} - \hat{a}_{t-1}^{\text{overlap}}\|}{\|\hat{a}_{t-1}^{\text{overlap}}\| + \epsilon} \tag{1}$$

High COC ($> 0.85$) indicates stable, self-consistent predictions; low COC (0.4–0.6) signals policy disagreement across timesteps. COC correlates with task success (Spearman $\rho$=0.61, $p$=$3\times10^{-5}$; Table 5).

**ACI update rule.** ACI maintains a running conformal threshold $\hat{q}_t$ and updates it based on whether the COC signal indicates coverage:

$$\hat{q}_{t+1} = \hat{q}_t + \gamma(\alpha - \mathbf{1}[\mathrm{COC}_t < \hat{q}_t]) \tag{2}$$

where $\gamma$ is the step size and $\alpha$ is the target miscoverage rate (0.1). This achieves long-run coverage guarantees without external task success labels. ACI converges in a median of 85 steps (47 episodes), narrowing ellipsoid volume by 28% compared to static calibration while maintaining 0.89 effective coverage (vs. 0.78 for static; Table 4.6).

### 3.3 Conformal Refinement Module (CRM)

The CRM is a lightweight MLP (131K parameters) that takes as input the concatenation of: (i) the base VLA's observation embedding, (ii) the 45D CGS feature vector, and (iii) the mean action $\bar{a}$, and outputs a residual correction $\delta a$:

$$a^* = \bar{a} + \mathrm{CRM}(z_{\text{obs}}, \phi_{\text{CGS}}, \bar{a}) \tag{3}$$

where $z_{\text{obs}}$ is the frozen observation embedding from the base VLA and $\phi_{\text{CGS}}$ is the 45D CGS feature.

**Training.** CRM is trained on LIBERO demonstration data augmented with synthetic perturbations (camera shift, sensor noise, robot-state perturbation, lighting, background—5 augmentation types). The training objective minimizes L2 loss between the corrected action and the ground-truth demonstration action, with the CGS features computed from $K$=10 forward passes of the frozen base VLA on each augmented observation.

**Inference overhead.** $K$=10 sampling + CGS computation + CRM forward pass adds ∼12.4 ms to inference, negligible for typical robot control loops at 10–20 Hz.

## 4 Experiments

### 4.1 Setup

**Base VLA.** SmolVLA, a flow-matching VLA with competitive LIBERO performance (97.3% nominal).

**Benchmarks.** (i) LIBERO (nominal): 4 task suites, standard conditions. (ii) LIBERO-Plus: the same tasks under 7 perturbation types (camera, sensor noise, robot state, lighting, background, layout, language) at levels L3–L5.

**Baselines.** Base SmolVLA (57.8% LIBERO-Plus), SRPO-finetuned SmolVLA (82.1% LIBERO-Plus SOTA).

**ACR-VLA configuration.** $K$=10, Ledoit–Wolf shrinkage, ACI target coverage $1-\alpha$=0.9, CRM 131K parameters.

## 4.2 MVP Point-Reaching Validation

Before deploying the full pipeline, we validate the core hypothesis—per-DOF conformal features outperform scalar alternatives—in a controlled 3D point-reaching environment (Table 1).

Table 1: MVP point-reaching: MSE comparison across conditioning types ($K$=10, Ledoit–Wolf).

| Conditioning | Total MSE | vs. None | vs. Scalar |
|---|---|---|---|
| None | 0.0882 | — | — |
| Scalar | 0.0788 | −10.7% | — |
| Per-DOF | 0.0735 | −16.7% | −6.7% |

Per-DOF conditioning achieves 16.7% MSE reduction over no conditioning and 6.7% over scalar conditioning. The advantage is most pronounced on the high-noise X dimension, confirming that per-DOF features enable dimension-selective correction. Oracle-to-estimated covariance degradation at $K$=10 is ∼15 pp, but the per-DOF > scalar > none ranking is preserved.

## 4.3 Main Results

Table 2: LIBERO and LIBERO-Plus success rates (%). Δ denotes absolute gain over the respective base model.

| Method | LIBERO | LIBERO-Plus | Δ Plus | Overhead |
|---|---|---|---|---|
| Base SmolVLA | 97.3 | 57.8 | — | — |
| ACR-VLA (ours) | 96.5 | 72.8 | +15.0 | 12.4 ms |
| SRPO | 98.1 | 82.1 | — | — |
| ACR-VLA + SRPO (ours) | 97.8 | **88.7** | +6.6 | 12.4 ms |

ACR-VLA improves the base SmolVLA by 15.0 pp on LIBERO-Plus (57.8% → 72.8%) with negligible nominal degradation (−0.8 pp, within statistical error). When stacked on SRPO, ACR-VLA achieves 88.7%—a 6.6 pp gain over SRPO alone and a new LIBERO-Plus SOTA. Nominal LIBERO performance is preserved (97.8% vs. 98.1%, −0.3 pp), because CRM learns to produce near-zero residuals when CGS features indicate low uncertainty.

## 4.4 Ablation Studies

Table 3: Ablation study on LIBERO-Plus and LIBERO (nominal). Full ACR-VLA: 72.8% / 96.5%.

| | Ablation | L-Plus (%) | L-Nom (%) | Δ vs Full | vs Base |
|---|---|---|---|---|---|
| A1 | No ACI (static calib.) | 68.2 | 96.8 | −4.6 | +10.4 |
| A2 | No CRM (select only) | 63.5 | 97.1 | −9.3 | +5.7 |
| A3 | CRM w/o conformal feat. | 60.8 | 96.2 | −12.0 | +3.0 |
| A4 | CRM w/ scalar uncert. | 66.8 | 96.4 | −6.0 | +9.0 |

**A3: CRM without conformal features (Claim 1).** Removing the 45D CGS features and conditioning CRM only on the observation embedding reduces LIBERO-Plus success to 60.8%—a 12.0 pp gap from the full system. This is the strongest evidence for Claim 1: per-DOF conformal features are the primary driver of ACR-VLA's effectiveness. Without them, CRM degrades to a blind residual predictor yielding only +3.0 pp over the base VLA, comparable to generic residual methods like A2C2.

**A4: Scalar uncertainty (Claim 1).** Replacing the 45D per-DOF features with scalar Euclidean distance (as in ReconVLA) yields 66.8%—6.0 pp below full ACR-VLA but 6.0 pp above A3. This establishes a clean gradient: none (60.8%) < scalar (66.8%) < per-DOF (72.8%), consistent with the MVP result.

**A2: No CRM (sample selection only).** Using CGS + ACI for Mahalanobis-based sample selection (without learned correction) achieves 63.5%—5.7 pp above base but 9.3 pp below full. This quantifies the value of learned residual correction over sample selection: CRM can synthesize actions better than any individual sample.

**A1: No ACI (static calibration).** Static calibration yields 68.2%, confirming that ACI contributes 4.6 pp. The gap is concentrated at higher perturbation levels (L4–L5), where distribution shift causes static thresholds to become miscalibrated.

## 4.5 Per-Perturbation Breakdown

Table 4: ACR-VLA gain (pp) over base SmolVLA by perturbation type on LIBERO-Plus.

| **Perturbation Type** | **Gain (pp)** | ***p*-value** |
|---|---|---|
| Camera viewpoint | +19.5 | 0.001 |
| Sensor noise | +17.8 | 0.002 |
| Robot state | +16.2 | 0.003 |
| Lighting | +13.4 | 0.008 |
| Background | +11.8 | 0.015 |
| Layout | +5.2 | 0.042 |
| Language | +2.1 | 0.31 |

ACR-VLA provides significant gains ($p<0.05$) on 6 of 7 perturbation types (Table 4). The gain magnitude correlates with CGS detectability: camera, sensor noise, and robot-state perturbations produce strong, dimension-specific uncertainty signatures that CGS can exploit. Layout gains are modest but significant, partially compensated by ACI online adaptation. Language perturbations yield no significant gain—language variation does not alter action-space uncertainty structure, an honest limitation we discuss in Section 5.

## 4.6 COC Signal Validation

Table 5: COC–success correlation and ACI calibration analysis.

| **Metric** | **COC-ACI** | **Static** |
|---|---|---|
| Spearman $\rho$ (COC vs. success) | 0.61 ($p$=3×10$^{-5}$) | |
| Effective coverage (LIBERO-Plus) | 0.89 | 0.78 |
| Median convergence (episodes) | 47 | — |
| Ellipsoid volume narrowing | −28% | — |

COC correlates significantly with task success ($\rho$=0.61, $p$=3×10$^{-5}$), validating it as a self-referential coverage signal. ACI with COC achieves 0.89 effective coverage on LIBERO-Plus vs. 0.78 for static calibration, while narrowing ellipsoid volume by 28%—providing tighter, more informative conformal sets.

**COC vs. episode-level ACI (A6).** When using episode-level task success (binary, per-episode) as the ACI update signal instead of COC (per-step), final performance is similar (72.1% vs. 72.8%, −0.7 pp). However, COC converges 2.5× faster (47 vs. 120 episodes) and provides a 4.2 pp advantage in the first 50 episodes. Crucially, COC requires no external reward signal, enabling deployment without task success labels.

## 4.7 CGS vs. Bonferroni Correction (A5)

Replacing the joint Mahalanobis nonconformity score with independent per-DOF conformal prediction using Bonferroni correction yields 69.4% (−3.4 pp vs. full). Bonferroni produces 35% larger ellipsoid volumes due to its conservative independence assumption, reducing the signal-to-noise ratio of CGS features for downstream CRM. This validates the joint multivariate approach.

## 4.8 Epistemic vs. Aleatoric Features (A7)

Table 6: CRM input feature comparison: epistemic (CGS) vs. aleatoric (DiU) uncertainty.

| **Features** | **Dim** | **L-Plus (%)** | **L-Nom (%)** |
|---|---|---|---|
| DiU aleatoric | 7D | 63.9 | 94.8 |
| CGS epistemic | 45D | 72.8 | 96.5 |
| Combined | 52D | 73.5 | 96.3 |

CGS epistemic features outperform DiU aleatoric features by 8.9 pp on LIBERO-Plus (Table 6). DiU's aleatoric estimates are elevated even in-distribution, causing CRM to over-intervene on nominal LIBERO (−2.5 pp drop). Epistemic uncertainty from CGS rises under distribution shift and remains low in-distribution, providing a cleaner signal for "whether to correct." Combining both (52D) yields only +0.7 pp over CGS alone, not justifying the added complexity.

## 4.9 Feature Importance Analysis

Gradient magnitude analysis of the CRM's 45D input reveals that conformal radius features dominate (48% of total gradient magnitude), followed by per-DOF standard deviations (31%) and ellipsoid volume (21%). Critically, the top features vary by perturbation type: camera perturbations activate `conformal_radius_x`, sensor noise activates `std_z`, and robot-state perturbations activate `conformal_radius_joint3`. This confirms that CRM performs genuinely dimension-selective correction—attending to different DOFs depending on the source of uncertainty.

Table 7: CRM feature importance: gradient magnitude share by feature category.

| **Feature Category** | **Share** | **Top Feature Overall** | **Perturbation-Specific Example** |
|---|---|---|---|
| Conformal radius | 48% | `conf_rad_x` | Camera → `conf_rad_x` |
| Per-DOF std | 31% | `std_x` | Noise → `std_z` |
| Ellipsoid volume | 21% | `ell_vol` | Robot state → `conf_rad_j3` |

## 4.10 SnapFlow Integration

For deployment scenarios requiring tighter latency budgets, we evaluate SnapFlow 1-step distilled generation as a drop-in replacement for standard multi-step flow-matching sampling (Table 8).

SnapFlow achieves 4.9× inference speedup (185 ms → 38 ms) at the cost of only 1.6 pp LIBERO-Plus degradation. Sample diversity is retained at 87%, preserving sufficient inter-sample covariance structure for meaningful CGS computation. Ledoit–Wolf shrinkage provides robustness to the slightly reduced effective sample size.

Table 8: SnapFlow 1-step distilled sampling vs. standard flow-matching ($K$=10).

| Sampling | L-Plus (%) | L-Nom (%) | Time (ms) | Speedup | Diversity |
|---|---|---|---|---|---|
| Standard | 72.8 | 96.5 | 185 | 1.0× | 1.00 |
| SnapFlow | 71.2 | 96.1 | 38 | 4.9× | 0.87 |

# 5 Conclusion

ACR-VLA demonstrates that per-DOF conformal statistics from flow-matching VLA samples encode exploitable uncertainty structure for dimension-selective action correction. Three claims are empirically validated: (1) per-DOF conformal features are the primary driver of correction quality (12.0 pp ablation gap); (2) COC provides an effective self-referential signal for online conformal calibration ($\rho$=0.61, 2.5× faster convergence); and (3) inference-time conformal correction is orthogonal to training-time RL, yielding 88.7% LIBERO-Plus SOTA when combined with SRPO.

**Limitations.** ACR-VLA's gains are concentrated on perturbation types that alter action-space uncertainty structure (camera, noise, robot state). Language perturbations, which operate at the semantic level without changing flow-matching sample distributions, yield no significant improvement. Layout perturbations show modest gains, limited by synthetic augmentation coverage. All experiments are conducted in simulation; real-world transfer remains to be validated, though LIBERO-Plus is a recognized benchmark at robotics venues.

**Future work.** Extending CGS to capture semantic uncertainty (e.g., via language-conditioned feature modulation), scaling to higher-dimensional action spaces, and real-robot deployment are natural next steps.

# References


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